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Surface modification of sized vegetal fibers through direct fluorination for eco-composites

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ABSTRACT

Natural fibers are frequently used as alternatives of glass fibers as polymer matrix reinforcement to form composite materials. However, their natural hydrophilicity prevents them from being easily compatible with hydrophobic polymers, which represent the majority of matrices. To solve this, direct fluorination of sized natural fibers (flax fibers sized with DGEBA) was performed. EDX, FTIR and ^{19}F NMR analysis evidenced the chemical grafting of fluorine on DGEBA structure and the ablation of the oxiranes rings on this molecule. Chemical modifications of DGEBA have induced a hydrophobic character of this layer, by reducing at 0 the polar component of surface tension. Thereby, treated fibers are supposed to be perfectly chemically compatible with the hydrophobic polymer matrices (*e.g.* polypropylene). Additionally, the fluorination time also allows the dispersive component of surface tension and the rugosity of fibers to be tailored in order to perfectly adjust these characteristics and fit with the polymer. Moreover, this chemical modification was achieved without altering the mechanical properties of fibers for short fluorination times.

1. Introduction

Vegetal fibers are materials produced and recycled naturally for millions of years. These fibers are more and more used as alternatives of glass fibers for polymer matrices reinforcement to form composites materials [1–5]. Indeed, those compounds allow to develop natural, local and renewable resources while lightening the cost and the overall weight of these materials, thanks to the specific properties of natural fibers, equivalent to glass fibers one. This specificity provides them to be frequently used in the automotive and aeronautical industry [1,4,6]. Besides, their use is more environmentally friendly than using synthetic fiber (like glass, carbon or aramid) and therefore, fits perfectly with the environmental issue of the XXIst century. Indeed, using natural fibers with a biodegradable and/or biobased polymer matrix allows “eco-composite” with a low environmental footprint to be prepared.

Despite this, one of the main difficulties in the use of these compounds as composite reinforcement is their hydrophily. Indeed, vegetal fibers are mainly composed of cellulose, lignin, and hemicelluloses which have a large number of hydroxyl and carboxyl groups in their chemical structure that provides to lignocellulosic fibers a high polarity with a polar component of surface tension γ_s^p of 26,0 mN/m (and a dispersive component γ_s^d of 34 mN/m) [7]. To obtain good filler/matrix interfacial properties, both materials must have close polar and dispersive components of surface tension. However, polymer matrices are mainly hydrophobic, with a very low γ_s^p (Table 1). Consequently, most of the time, a poor interfacial adhesion is created between polymer matrix and vegetal fibers which leads to a decrease of the global mechanical performance of the composite [8–11]

In order to reduce the polar component of natural fiber surface tension, many different methods have been developed. As examples, corona treatment, torrefaction, acetylation, malleated coupling, mercerization, and peroxide treatments may be cited as well as a lot of other chemical and/or physical processes [19–24]. However, these processes are unfortunately costly, both time and energy consuming and may degrade the fibers (and the mechanical properties in the same way), and/or harmful to the environment and people by using toxic solvents. Another methods consist to carry out a chemical grafting on the fiber surface to provide good compatibility with a hydrophobic polymer (*e.g.* Polypropylene which is one of the most popular composite matrices). One of the representative examples of this technique is the use of maleic anhydride-grafted polypropylene (MAPP) to create a chemical “bridge” between fibers (natural or artificial) or the fiber

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